OOPs LAB
Spring Semester 2026
Assignment 8

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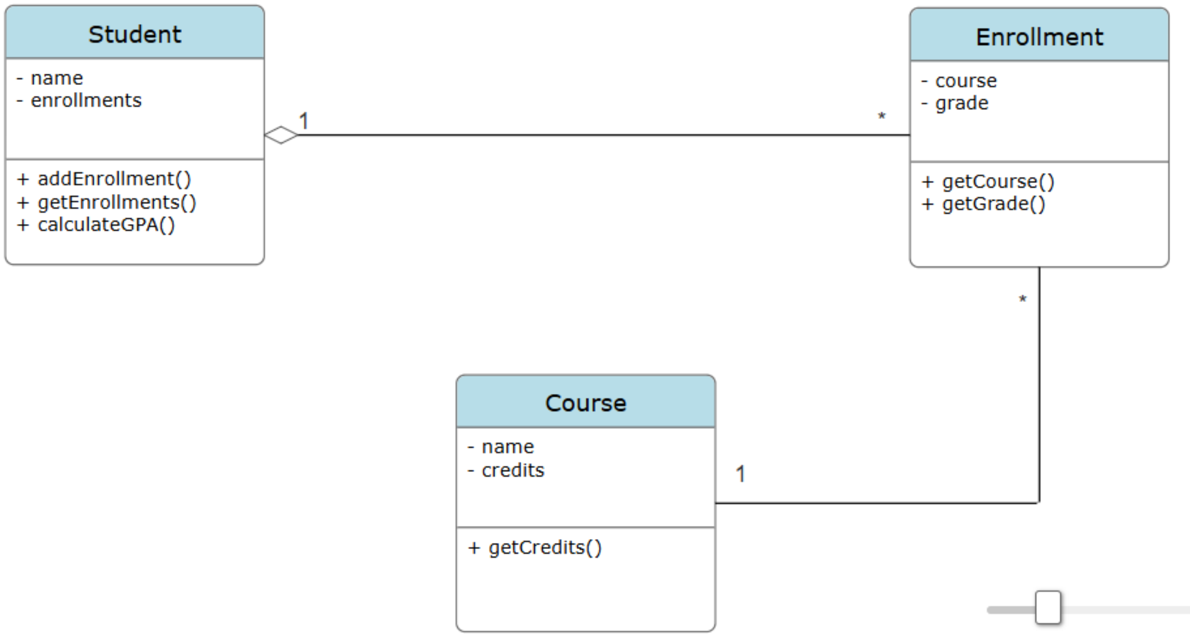
<u>Section 1 –</u>

Q1) UML Diagram

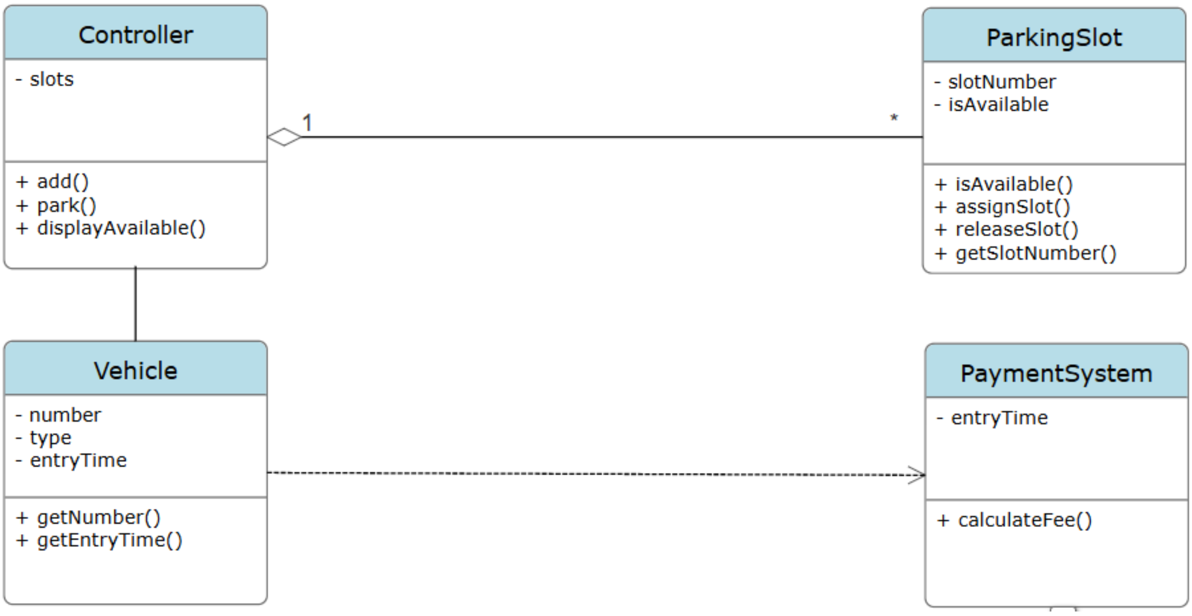


Here we are using the concept of **Expert Pattern** *Student* is the expert Class here and hence, *GPAUtil* is removed (entirely) and instead *computeGPA* is put into *Student* since it has all the information to compute the GPA i.e. all Courses grade and credits. This reduces coupling between *Student* and *GPAUtil*.

Q2) UML Diagram

Creator Pattern is used here. Putting *addPackage()* inside *Mission* **reduces coupling** between *Main* and *Package.* Since, Main no longer directly calls to *Package* also *Mission* and *Package* were already previously coupled, so it doesn't affect coupling between them.

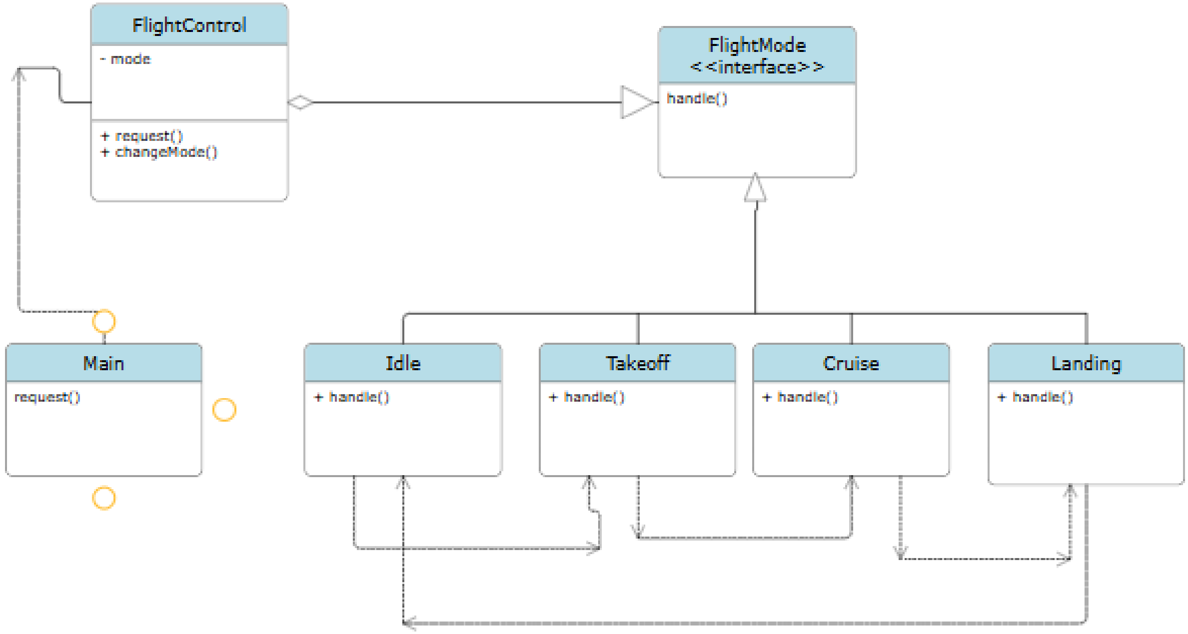
Q3) UML Diagram



Class *Controller* uses **Controller Pattern** and handles all the UI operations such as display and distribution tasks upon vehicle parking. Reduces **coupling** since now *Main* now longer needs to use *ParkingSlot* which is handled inside *Controller.* Also since *PaymentSystem* is the **Expert class** it handles the *calculateFee()* without interference from *Main.*

Section 2 –

UML Diagram:



The given FlightControl mechanism is using **State Design Pattern.** FlightMode can be in 4 different states – *Idle, Takeoff, Cruise and Landing.* Each of these classes use the same Interface *FlightMode* which implements function *handle()* and then shifts the state (FlightMode) to the next state as shown in the UML diagram.